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Breakage mechanics—Part I: Theory

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Abstract

Different measures have been suggested for quantifying the amount of fragmentation in randomly compacted crushable aggregates. A most effective and popular measure is to adopt variants of Hardin's [1985. Crushing of soil particles. J. Geotech. Eng. ASCE 111(10), 1177–1192] definition of relative breakage 'B_r'. In this paper we further develop the concept of breakage to formulate a new continuum mechanics theory for crushable granular materials based on statistical and thermo-mechanical principles. Analogous to the damage internal variable 'D' which is used in continuum damage mechanics (CDM), here the breakage internal variable 'B' is adopted. This internal variable represents a particular form of the relative breakage 'B_r' and measures the relative distance of the current grain size distribution from the initial and ultimate distributions. Similar to 'D', 'B' varies from zero to one and describes processes of micro-fractures and the growth of surface area. However, unlike damage that is most suitable to tensioned solid-like materials, the breakage is aimed towards compressed granular matter. While damage effectively represents the opening of micro-cavities and cracks, breakage represents comminution of particles. We term the new theory continuum breakage mechanics (CBM), reflecting the analogy with CDM. A focus is given to developing fundamental concepts and postulates, and identifying the physical meaning of the various variables. In this part of the paper we limit the study to describe an ideal dissipative process that includes breakage without plasticity. Plastic strains are essential, however, in representing aspects that relate to frictional dissipation, and this is covered in Part II of this paper together with model examples.
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1. Introduction

The mechanics of particle crushing, or particle breakage, is one of the most intractable problems in geosciences. The topic is of interest to many research disciplines including powder technology, minerals and mining engineering, geology, geophysics and geomechanics. Prior studies have normally been aimed towards characterising the evolution of the particle size distribution with increase in energy. Another motivation, which is of particular interest in the field of geomechanics, is to link between the crushing of particles and the mechanical response of the soil through behavioural constitutive models. Existing constitutive models are based on simple curve-fitting parameters, which are taken in isolation from unique stress–strain tests, following paths appropriate to the case in hand. The result is that geotechnical engineers often distrust the use of modern constitutive models, which presents one of the most pressing problems of soil mechanics today (Bolton, 2000). For example, none of the current soil mechanics constitutive models takes into account the effect of the grain size distribution, although this distribution is routinely measured at almost every engineering site. The ability of models to represent the evolution of this distribution during the crushing of particles under any mechanical boundary conditions, may potentially form the physical foundations for clarifying many geo-phenomena, and improve the confidence of geotechnical engineers in using constitutive models for sands.

A practical way of introducing the vulnerability of a collection of particles to crush is by constructing an enriched continuum model that mesoscopically averages the micro-crushing events. An exciting example was recently proposed by McDowell et al. (1996). Their theory aimed to explain the reasons behind isotropic hardening in critical state soil mechanics. By adopting a new work equation that includes dissipation from the fracture and creation of new surface area, their proposition was to use the term 'clastic hardening', rather than 'isotropic hardening', to highlight that it is the fragmentation process that causes the hardening phenomenon. In the particular case of one-dimensional compression, the fracture term may be degenerated and be substituted by a useful plastic dissipation term. When more general loading conditions apply, their approach converts to a typical plasticity formulation, essentially omitting the fact that plastic and fracture dissipations are separate. Consequently, their theory was limited to providing physical justification to the choice of a value for the hardening parameter λ in critical state soil models, rather than developing new models. Another interesting approach was undertaken by Ueng and Chen (2000), associating the energy consumption from crushing linearly with the increase of surface area. Unlike the clastic hardening approach that was aimed for one-dimensional compression, Ueng and Chen's approach was concerned only with shear deformations. Indraratna and Salim (2002) have later extended Ueng and Chen's approach by linking the energy consumption with Marsal's (1973) breakage index rather than the increase of surface area. This enabled Indraratna and Salim to extend the analysis for triaxial loading conditions. However, this required adding phenomenological parameters from curve-fitting triaxial stress–strain tests.

Another approach to understand the collective fragmentation process of an aggregate could be given by tracking the changes in the particle size distribution. Hardin (1985) underpinned the necessity for an adequate measure of the crushing in establishing continuum stress–strain models:

> *In order to understand the physics of the strength and stress–strain behaviour of soils and to devise mathematical models that adequately represent such behaviour, it is*

important to define the degree to which the particles of an element of soil are crushed or broken during loading.

By advancing the earlier works of Lee and Farhoomand (1967) and Marsal (1973), Hardin developed the concept of relative breakage. Although the relative breakage concept has been widely applied to '*define the degree to which the particles of an element of soil are crushed or broken during loading*' (e.g., Coop and Lee, 1993; Lee and Coop, 1995; Nakata et al., 1999), the issue of creating '*mathematical models that adequately represent such behaviour*' by rigorously incorporating the relative breakage concept has yet to be satisfactorily addressed.

The purpose of this paper is to establish a soundly based continuum theory that incorporates the concept of breakage. The pressing problems of the lack of physical meaning of the various parameters is addressed by consistently incorporating the evolution of grain size distribution in constitutive models by developing the concept of breakage.

2. Breakage

In this paper, we employ the concept of relative breakage and use it as an internal variable in a continuum mechanics formulation of constitutive models. The concept of relative breakage, or *breakage*, which varies from zero to one with changes in grain size distribution is different from the concept of *damage* which also varies from zero to one with changes in the amount of micro- voids and cracks. Their physical interpretation and role are totally different, if not opposite, as discussed in this section. Before considering breakage the concept of damage and its inadequacy for modelling grain crushing will be discussed.

2.1. The inadequacy of damage for modelling grain crushing

At the micro-scale level, damage may be interpreted as the creation of additional micro-surfaces of discontinuity surfaces: breaking of atomic bonds and enlargement of micro-cavities. Despite the discontinuous nature of such processes, the damage concept represents them in an effective continuous way by employing the framework of continuum mechanics. The mechanics of this particular continuum are called continuum damage mechanics (CDM). In this area of mechanics, a set of effectiveness rules is amended by adopting thermomechanical principles, and using damage as an internal variable. The scalar damage concept was first introduced by Kachanov (1958) and since then has become very popular in describing the deterioration of stiffness and strength in solid-like materials. The simplest definition of damage is given by

$$D = \frac{A_v}{A}, \tag{1}$$

where A is the total cross-section area of a surface within the unit cell oriented in any selected direction; A_v is the void matrix area within A. Since the void matrix area, or volume, can never exceed the total matrix area, the damage is bounded by

$$1 \geqslant D \geqslant 0, \tag{2}$$

where $D = 0$ denotes undamaged material and $D = 1$ represents complete damage. In a continuum mechanics sense, the damage scalar effectively describes the propagation of

micro-defects over the mesoscale volume element, by assuming that defects are isotropically distributed.

Generally, CDM models are *appropriate for solid-like materials*, which may undergo opening of micro-cracks and voids when extended (precursor to the formation of mesoscale fractures). During damage growth the effective elastic and strength moduli are experiencing an on-going deterioration. However, CDM models are *inappropriate for* discrete crushable *granular materials*. Firstly, because the definition in Eq. (1) becomes ambiguous. Secondly, because the surface area of the grains will not grow in extension as in an ideal 'damaged material'. Thirdly, because much more than solid-like materials, the surface area of the grains of brittle granular materials will grow under compression due to the crushing process. Fourthly, because the fracturing of the grains in compression is expected to increase the effective strength parameters, rather than decrease, due to the continuing reduction in the size of the particles. For these reasons, we seek an alternative representation of the internal changes via the relative breakage.

2.2. Definition of relative breakage

To quantify effectively the amount of fragmentation within the representative volume of randomly compacted crushable granular materials Hardin (1985) suggested the use of a relative breakage property based on the relative position of the current cumulative distribution from

(a) an initial cumulative distribution, and
(b) an arbitrary cut-off value of 'silt' particle size (of 0.074 mm).

The use of the latter implied that all particles, no matter what their original size is, will eventually become finer than this arbitrary cut-off value.

According to Hardin (1985):

> *Under extremely high stresses it would in principle be possible for all of the particles in a sample of soil to be crushed to the extent that no particles remain larger than 0.074 mm.*

However, this statement does not agree with the growing understanding that the grain size distribution of compacted aggregates should be bounded by an ultimate distribution, attained under extremely large confining pressure and extensive shear strains. For example, Turcotte (1986) argued that any initial distribution of particles will tend towards a self-similar distribution. This conclusion was supported by analysis of geological fault gouges by Sammis et al. (1987), which is possibly as close as one can get to the sufficient conditions for the generation of an ultimate grain size distribution. The idea that grain size distribution tends to be fractal is becoming more widely used. For example, McDowell and Bolton (1998) state that:

> *If particles fracture such that the smallest particles are in geometrically self-similar configurations under increasing macroscopic stress, with a constant probability of fracture, a fractal geometry evolves with the successive fracture of the smallest grains, in agreement with the available data.*

In the fragmentation process larger particles get cushioned by surrounding smaller particles, making them more resistive to crushing, so that smaller particles are more likely to be crushed. This cushioning effect was observed numerically and experimentally by Tsoungui et al. (1999). Fig. 1 presents their experimental result, indicating that larger particles get cushioned while neighbouring smaller particles continue to crush. It is therefore the constraining topological effects that control the form of the ultimate grain size distribution. This effect was used by Sammis et al. (1987) in proposing a unique confined organisation of particles that was proven to infer a fractal distribution if no further crushing was to emerge. Hence, this system may be seen in thermodynamic equilibrium and maximum entropy, because the geometry eliminates the occurrence of stress concentrations, or in other words because the energy is dispersed among the largest possible number of particles (which could be seen as thermodynamic micro-states). However, it is important to note that taking grains out of a sample with an ultimate fractal distribution, then reconstituting and reloading them, may lead to new fragments emerging because the 'old' thermodynamically equilibrated geometry has been lost. This would shift the previous ultimate fractal distribution to a new ultimate distribution (see experiments in Coop, 2006). This does not infer a reduction in entropy, or violation of energy conservation, but is simply because new external energy has been added for mixing the system. Therefore, the exact form of the ultimate distribution is still an open question, which would not necessarily be fractal, although a fractal model with a fractal dimension α of around 2.5–2.6 (Sammis et al., 1987) should be useful for most practical cases. Appendix A presents the derivation of the fractal grain size distribution by mass, but for generality we proceed with any plausible $F_u(d)$.

We therefore propose to adjust the original definition of the relative breakage by Hardin to weigh from zero to one the relative proximity of the current grain size cumulative distribution from

(a) an initial cumulative distribution, and
(b) an ultimate cumulative distribution.

This definition is presented in Fig. 2. Therefore, three different grain size cumulative functions define our breakage definition: the current 'F', initial 'F_0', and ultimate 'F_u'

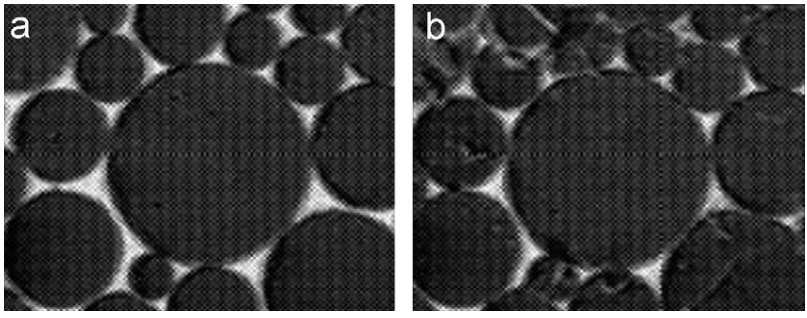


Fig. 1. Visualisation of the 'cushioning effect' in disks assembly via experiment (after Tsoungui et al., 1999, colour inverted from original figure).

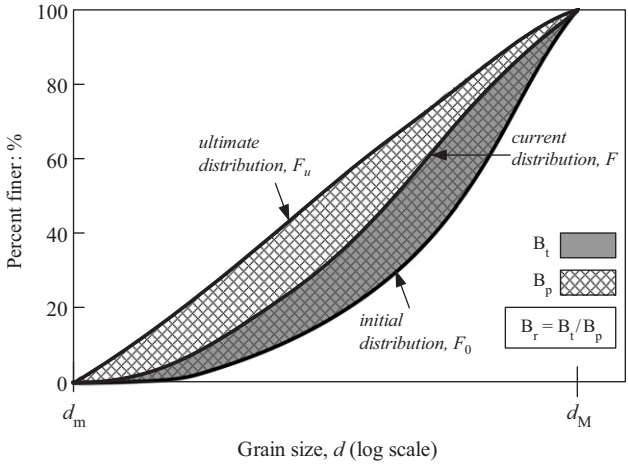


Fig. 2. Modified definition of Hardin's breakage index B_r. It is possible to define a universal initial cumulative distribution as a vertical line through d_M (as discussed with Randolph, 2006, and as concurrently proposed by Wood, 2006). In this way, B_r would also be spanned from zero to unity, although most samples would actually start at an intermediate value of B_r. It should be noted, however, that the ultimate distribution is a function of the initial grading (Coop, 2006). This is why for generality we adopt the current definition of B_r (that is zero when $F(d) = F_0(d)$). The universal B_r concept may be easily followed by taking $F_0(d)$ as the heaviside function around d_M and starting with an initial B_r value.

cumulative functions

$$F(d) \equiv F(\Delta < d) = \int_{d_m}^{d} p(\Delta)\,\mathrm{d}\Delta,$$
$$F_0(d) \equiv F_0(\Delta < d) = \int_{d_m}^{d} p_0(\Delta)\,\mathrm{d}\Delta,$$
$$F_u(d) \equiv F_u(\Delta < d) = \int_{d_m}^{d} p_u(\Delta)\,\mathrm{d}\Delta, \tag{3}$$

where Δ denotes the particles of size less than d; $p(\Delta)$, $p_0(\Delta)$, and $p_u(\Delta)$ denote the current, initial and ultimate grain size distributions; $p(\Delta)\,\mathrm{d}\Delta$, $p_0(\Delta)\,\mathrm{d}\Delta$, and $p_u(\Delta)\,\mathrm{d}\Delta$ denote the current, initial and ultimate probabilities of a particle to exist in the fraction $\mathrm{d}\Delta$; d_m is the smallest particle size, as opposed to d_M which will denote the largest particle size.

In Fig. 2, the relative breakage is defined as an area ratio:

$$B_r = \frac{B_t}{B_p}. \tag{4}$$

The 'breakage potential' is defined by integrating the entire area in Fig. 3(a) over the 'y-axis' (where y corresponds to the fraction relative finer value from 0 to 1, instead of 0–100%):

$$B_p = \int_0^1 \log\frac{d_0(y)}{d_u(y)}\,\mathrm{d}y. \tag{5}$$

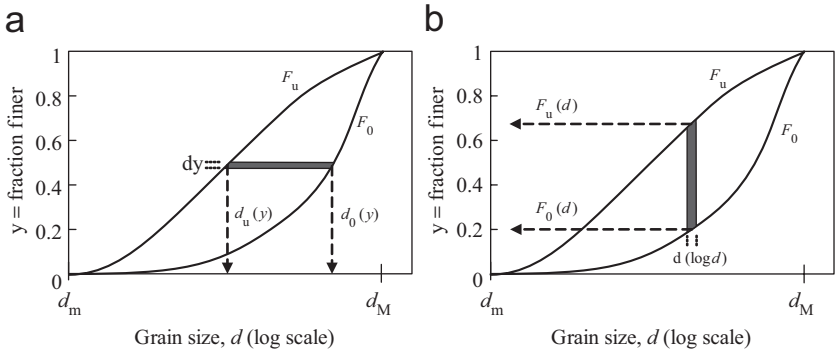


Fig. 3. Two ways to integrate the area associated with B_p: (a) along the y-axis (following Hardin, 1985), or (b) along the logarithmic x-axis (i.e., over $\log d$).

This equation follows the way Hardin integrated the area, where the constant value of the silt size 0.074 mm which Hardin used in the denominator in his original definition is replaced with d_u, denoting the ultimate grain size as a function of the given y value.

An alternative way to obtain B_p is to integrate the area over the $\log d$ scale (see Fig. 3(b)):

$$B_p = \int_{d_m}^{d_M} (F_u(d) - F_0(d))\,\mathrm{d}(\log d) = \ln^{-1}10 \int_{d_m}^{d_M} (F_u(d) - F_0(d))d^{-1}\,\mathrm{d}d \tag{6}$$

noting that $\mathrm{d}(\log d) = d^{-1}\ln^{-1}10\,\mathrm{d}d$. This form of integration is useful to our purpose.

For the same reasons, Hardin's 'total breakage' definition may be represented by

$$B_t = \ln^{-1}10 \int_{d_m}^{d_M} (F(d) - F_0(d))d^{-1}\,\mathrm{d}d \tag{7}$$

to underline the amount of crushing that the sand has undergone. It is interesting to note that in information technology, the value B_t represents the so-called Kick's grinding energy (e.g., Zhukov et al., 1998) normalised by the energy constant C_k.

The exact definition of relative breakage follows by combining Eqs. (4), (6) and (7):

$$B_r = \frac{\int_{d_m}^{d_M} (F(d) - F_0(d))d^{-1}\,\mathrm{d}d}{\int_{d_m}^{d_M} (F_u(d) - F_0(d))d^{-1}\,\mathrm{d}d}. \tag{8}$$

Given the ultimate grain size cumulative distribution, the relative breakage is limited by

$$1 \geqslant B_r \geqslant 0, \tag{9}$$

where $B_r = 0$ denotes unbroken material and $B_r = 1$ represents complete breakage.

2.3. Fractional breakage

Let us define the fractional breakage $B(d)$, accounting for the breakage of the different fractions. Assume that this measure is fractionally independent, i.e., being identical for all of the particle fractions and satisfying $B(d) = B$. Further assume that B is given

corresponding to Eq. (8):

$$B = \frac{p_0(d) - p(d)}{p_0(d) - p_u(d)}. \tag{10}$$

The effectiveness of this working hypothesis is comparable to the use of an isotropic damage scalar in CDM models. Although it is quite clear that damage is generally a fourth order tensor (Simo and Ju, 1987), the isotropic scalar has provided the starting point to development of CDM models. Any course on damage mechanics would almost always start by portraying the scalar representation (e.g., Lemaitre, 1992), and the use of this concept allowed to establish consistent coupled damage plasticity models (Einav et al., 2006).

The assumption of fractional independency is also equivalent to the use of a single plastic strain variable in elastoplastic models for granular media. It is quite clear that the single plastic strain variable represents in an effective manner the much richer and realistic description of the relative micro-sliding and rotation between the many contacting particles in the representative volume element. Nevertheless, the use of the simple single plastic strain measure has proven useful.

By rearrangement and integration:

$$p(d) = p_0(d)(1 - B) + p_u(d)B, \tag{11}$$

$$F(d) = F_0(d)(1 - B) + F_u(d)B. \tag{12}$$

Therefore, for a given B the current grain size distribution and grain size cumulative functions could be assessed at any time. It is easy to see that the fractional breakage definition of B satisfies the integrated cumulative definition of the relative breakage B_r, simply by multiplying Eq. (12) by d^{-1} and integrating once more:

$$B = \frac{\int_{d_m}^{d_M} (F_0(d) - F(d)) d^{-1}\, \mathrm{d}d}{\int_{d_m}^{d_M} (F_0(d) - F_u(d)) d^{-1}\, \mathrm{d}d} = B_r. \tag{13}$$

Therefore, by accepting the fractional independency of breakage on particle sizes, the value of B may be evaluated effectively by measuring experimentally the relative breakage B_r. In the classical statistical mechanical theory of discrete systems, the properties of mesoscopic bulk matter are deduced from the microscopic elemental properties once the grain size distribution is defined for the microscopic property (e.g., Shahinpoor, 1980). Since the assumption of fractional independency of breakage links the relative breakage to the grain size distribution (as well as to the volume of the cumulative functions) it could be integrated within a thermomechanical continuum formulation as we will show later. Thus, from this point onwards, we shall adopt the symbol B, and the term breakage, to designate the fractional breakage as well as the relative breakage for conciseness.

The effectiveness of the fractional identity hypothesis is now examined via the use of the following example.

2.4. Example of breakage measurement

In this section we explore the recent results presented by Coop et al. (2004) which applied extensive shear strains on an initially uniformly graded Dog's Bay sand in the ring

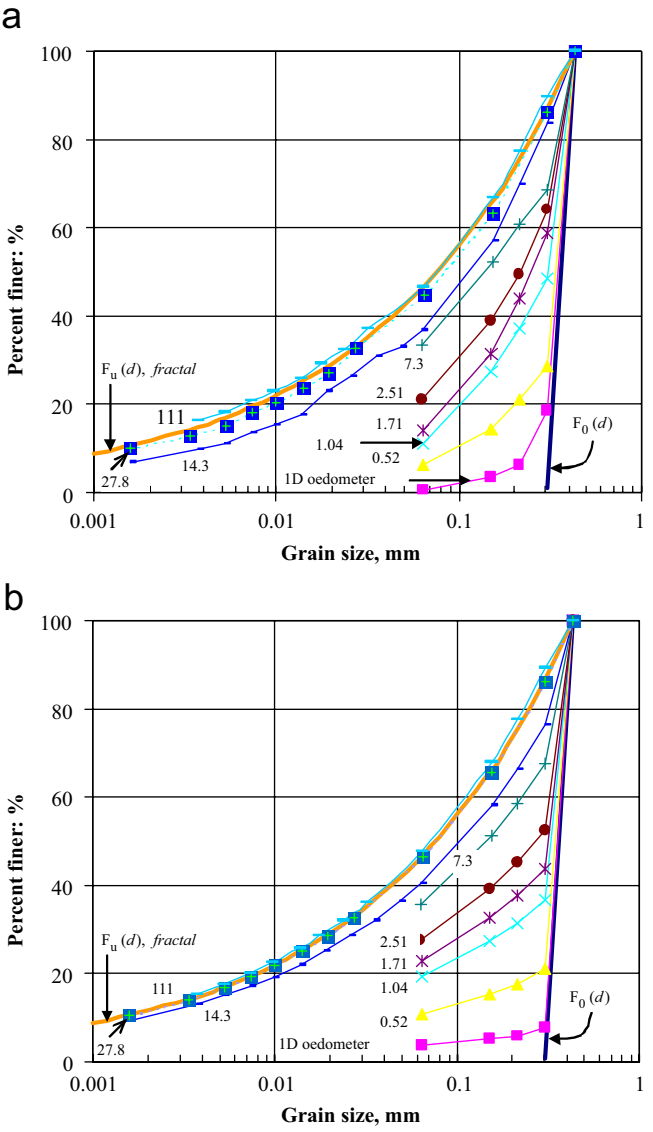


Fig. 4. Evolution of grain size distribution of Dog's Bay sand during shear tests (notional values of shear strains are indicated next to the curves), following one-dimensional compression in the oedometer to a vertical stress of 800 kPa: (a) presents the original data by Coop et al. (2004), with additional ultimate distribution F_u. (b) presents theoretical curves by taking the fractional identity working hypothesis.

shear apparatus, following one-dimensional compression to a vertical stress of 800 kPa. Fig. 4(a) presents the evolution of grain size distribution in these tests, with the notional of shear strain values indicated next to the curves.

Based on Hardin's definition of breakage, Coop et al. (2004) measured the breakage as a function of the applied shear strain (defined as the revolution displacement of the shear

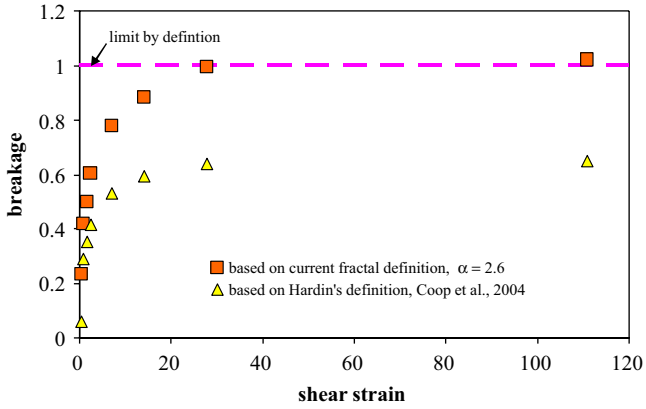


Fig. 5. Breakage measured based on the original definition by Hardin and based on the current definition using an ultimate fractal distribution with $\alpha = 2.6$ (i.e., $F_u(d) = (d/d_M)^{0.4}$).

rings divided by a hypothetic shear band height), and this is indicated in Fig. 5. The curve is asymptotic, but not towards unity as originally envisaged by Hardin in proposing his definition.

Let us start by examining our modification to Hardin's breakage definition, replacing the cut-off silt grain size curve by an assumed ultimate distribution. As mentioned earlier, the exact form of distribution is unknown, but the fractal distribution should be a good starting point. For that reason, let us assume that the ultimate distribution is indeed fractal, using Eq. (A.8):

$$F_u(d) = \left(\frac{d}{d_M}\right)^{3-\alpha}. \tag{14}$$

Coop et al. (2004) have noted a fractal dimension of 2.6, within the suggested theoretical range of values of Sammis et al. (1987), by plotting the results in a log–log space and linking the fractal dimension to the slope of the line in this space. This particular fractal distribution could, in fact, be added to the original semi-log plot in Fig. 4 by using $F_u(d) = (d/d_M)^{0.4}$ based on Eq. (14).

For the same shear strain values, the breakage values could be recalculated based on our modification to Hardin's proposition. This set of results is added in Fig. 5, showing how the asymptote now tends towards unity.

To examine the capabilities of the fractional identity working hypothesis, let us assume that the only known parameters, at this stage, are the initial and ultimate distributions. This would be the common practical situation. For a given breakage value we can deduce a theoretical cumulative distribution via Eq. (12), and this is represented in Fig. 4(b). Of course, since those are theoretical distributions, they are smooth, unlike the experimental curves in Fig. 4(a). Notably, the fractional identity working hypothesis seems to be extremely effective in capturing the progression of the curves with the increase in breakage, in particular when considering the minimal a priori information used.

3. Statistical homogenisation

Since according to Eq. (11) the breakage directly affects the shape of the current grain size distribution, it can be incorporated in constitutive models if they become dependent on the grain size distribution. It is well recognised that the particle size distribution is one of the main properties that influence the constitutive behaviour of granular materials. However, at least to the author's knowledge, the use of particle size distributions has never been integrated directly in any of the constitutive relations that geotechnical researchers have used in the past. How then should the grain size distribution be introduced consistently within a continuum mechanics formulation?

The approach adopted here is to use statistical homogenisation, via statistical averages, which is a known strategy in statistical mechanics (e.g., Shahinpoor, 1980; Bagi, 2003; Goddard, 2004).

3.1. Statistical average

The stress–strain behaviour of sand analysed in elementary laboratory tests (e.g., triaxial, simple shear, etc.) will not be repetitive unless the characteristic size of the individual particles is significantly small compared to the overall laboratory sample dimensions. However, the common practise in the laboratory is to restrict the tests for sands that have infinitesimally small particles compared to the sample dimensions. This condition ensures that the results will become essentially repetitive for the same sand even if it is being tested several times. This characterises the spatial ergodicity property of the material, i.e., the independency of the material behaviour from the spatial organisation of the particles (Harr, 1977).

In this case we may use statistical homogenisation, implementing the grain size distribution as a weighting average function over the microscopic variables. In general, the average $\overline{A} \equiv \langle A \rangle$ of the microscopic variable $A(d)$, which represents all of the grains with the size d, is given by

$$\overline{A} \equiv \langle A \rangle = \int_{d_m}^{d_M} A(d) p(d)\, \mathrm{d}d. \tag{15}$$

Noting the use of Eq. (11), in our formalism we get

$$\overline{A} \equiv \langle A \rangle = \langle A \rangle_0 (1 - B) + \langle A \rangle_u B, \tag{16}$$

where we define two additional averages, based on the initial and ultimate grain size distributions:

$$A_0 \equiv \langle A \rangle_0 = \int_{d_m}^{d_M} A(d) p_0(d)\, \mathrm{d}d,$$
$$A_u \equiv \langle A \rangle_u = \int_{d_m}^{d_M} A(d) p_u(d)\, \mathrm{d}d. \tag{17}$$

Any microscopic variable 'A' at the grain size level d, may be treated using Eq. (16). In initial stages, before the onset of breakage we note that $\overline{A} \equiv \langle A \rangle = \langle A \rangle_0$.

For illustration, the property $A(d)$ may be given as the stored energy within notional grains with the size d, i.e., $A(d) = \psi(d)$, and this will be used later on. As another example,

$A(d)$ could simply be the grain size d itself, i.e., $A(d) = d$. In this case the average particle size is expressed by

$$\overline{d} \equiv \langle d \rangle = \int_{d_m}^{d_M} dp(d)\,\mathrm{d}d \tag{18}$$

which is simply the first order moment J_1 of the grain size distribution $p(d)$. For later considerations, the second order moment of the grain size distribution $p(d)$ is given by

$$J_2 \equiv \langle d^2 \rangle = \int_{d_m}^{d_M} d^2 p(d)\,\mathrm{d}d \tag{19}$$

noting that the second order moment describes the way in which the grain sizes are distributed about the mean grain size $\overline{d}$.

Most generally, the nth order moment of the grain size distribution $p(d)$ is

$$J_n \equiv \langle d^n \rangle = \int_{d_m}^{d_M} d^n p(d)\,\mathrm{d}d. \tag{20}$$

4. Elastic continuum breakage mechanics theory

We are now able to integrate the concept of breakage within modern thermomechanical procedures and complete the basic foundations of the new theory.

4.1. General thermodynamics statement

A convenient form to represent the two laws of thermodynamics of rate independent compacted agglomerate in isothermal conditions is stated as follows (e.g., Walsh and Tordesillas, 2004; Collins and Einav, 2005):

$$\tilde{W} = \delta\Psi + \tilde{\Phi}, \quad \tilde{\Phi} \geqslant 0, \tag{21}$$

where Ψ and $\delta\Psi$ are the Helmholtz free energy and its increment; $\tilde{\Phi}$ is the non-negative (increment of) energy dissipation; $\tilde{W}$ is the (increment of) mechanical work done on the RVE boundaries. The use of the tilde symbol over $\tilde{W}$ and $\tilde{\Phi}$ represents an increment, deliberately different from the proper notation of an increment δ. This is made specifically to highlight that while Ψ is a state function, which therefore has a proper differential, W and Φ are not and only their increments could be defined. The increment of work is expressed by

$$\tilde{W} = \sigma : \delta e, \tag{22}$$

where σ and e are the stress and strain tensors that applies to the boundaries of the RVE, while δe is the increment of the strain. By accepting additivity of internal energy and rate of dissipation, we may express the Helmholtz free energy, for example, via Eq. (17):

$$\Psi \equiv \langle \hat{\psi} \rangle = \langle \hat{\psi} \rangle_0 (1 - B) + \langle \hat{\psi} \rangle_u B = \Psi_0 (1 - B) + \Psi_u B, \tag{23}$$

where $\hat{\psi}$ denote the density distributions of the Helmholtz free energy in the different particle size fractions d. Those grains are being treated as elastic up to failure.

4.2. Elastic CBM models

Under the action of external loads, compacted granular materials may undergo several dissipative mechanisms at the micro-level. Those mechanisms generally include dissipation from the fracturing of the particles (here described via breakage), and from their relative frictional sliding and spin. However, when the compacted aggregate is isotropically compressed, the frictional modes are often triggered only after breakage and fracture. In other words, the particles are interlocked with no degrees of freedom allowed for rotation and slippage, at least until the breakage begins. Subsequently, the rearrangement of the fragments may be accompanied with further dissipation due to friction, in the general sense of plasticity. Therefore, in these conditions, the plasticity is only secondary to breakage. Hence, developing models that account for the breakage, without considering plastic straining, is a good starting point before proceeding to the full analysis. In this paper we limit the study to general elastic CBM models, ignoring frictional dissipation for the time being. This component is dealt with in Part II of the paper (Einav, 2006).

Our expectation is that the average stored energy within grains of a given size d_1 would be different to the average stored energy within grains of another size d_2. Let us assume that the difference between their stored energy is relative to their size. For this purpose we postulate the following Helmholtz free energy density function in particles with the size d:

$$\hat{\psi} = \hat{\psi}(d, e) = f_{\psi}(d)\psi_r(e), \tag{24}$$

where e is the gross element strain, and the 'energy split function' $f_{\psi}(d)$ plays the role of 'splitting' the stored energy between the different particle size fractions. For a reference particle size d_r, we impose $\hat{\psi} = \psi_r(e)$, so that

$$f_{\psi}(d_r) = 1. \tag{25}$$

The statement in Eq. (24) is referred to as the 'hypothesis of energy split', and will be tested later in Section 4.3.

Rewriting Eq. (23) using (24) gives

$$\Psi \equiv \psi_r(e)[(1 - B)m_0 + Bm_u], \tag{26}$$

where we identify m_0 and m_u

$$m_0 = \int_{d_m}^{d_M} f_{\psi}(\Delta)p_0(\Delta)\,\mathrm{d}\Delta = \langle f_{\psi}(d)\rangle_0, \tag{27}$$

$$m_u = \int_{d_m}^{d_M} f_{\psi}(\Delta)p_u(\Delta)\,\mathrm{d}\Delta = \langle f_{\psi}(d)\rangle_u \tag{28}$$

as two—physical—model parameters, which relate to the initial and ultimate particle size distributions via the energy split function $f_{\psi}(d)$. It is interesting to note that the stored energy function in the reference grain size may be expressed as

$$\psi_r(e) = \frac{\Psi(B \equiv 0)}{m_0}, \tag{29}$$

i.e., the average value of the Helmholtz free energy, before any breakage occurs, normalised by the parameter m_0. This conclusion suggests that before the initiation of

breakage, the energy stored in the reference grains is simply the overall stored energy divided by m_0.

The increment of the statistical average of the Helmholtz free energy is then

$$\delta\Psi \equiv \frac{\partial\psi_r(e)}{\partial e}[(1-B)m_0 + Bm_u]\delta e + \psi_r(e)(m_u - m_0)\delta B. \tag{30}$$

Unlike the elastic stored energy, which is separable and scalable between the fractions, the dissipation due to breakage is not because the crushing of an individual particle effects, from a probabilistic point of view, to the entire particle size distribution below the size of the crushed particle. In information technology, this process is described via the use of the transition function (e.g., Zhukov et al., 1998). Without denoting the contribution from the various fractions, the breakage dissipation for rate independent granular materials is assumed simply as a first order homogeneous function of the breakage rate of change. This assumption is equivalent to the one taken in conventional rate-independent plasticity. McDowell and Khan (2003) have recently observed that grain size distribution may in fact be slightly shifting due to creep crushing. To take such rate effects into account, viscosity may be considered by releasing the first order homogeneity condition and treating the theory as viscous-breakage, equivalent to viscous-plasticity. Considering Euler's theorem for this type of homogeneous functions, we write

$$\tilde{\Phi}_B = \tilde{\Phi}_B(B, \delta B) = E_B \delta B \geqslant 0, \tag{31}$$

where we designate

$$E_B = \frac{\partial\tilde{\Phi}_B(B, \delta B)}{\partial(\delta B)}. \tag{32}$$

The physical meaning of this measure will soon be revealed. We note, from Eq. (31), that for the breakage to grow (i.e., for $\delta B \geqslant 0$), we should expect to have

$$E_B \geqslant 0. \tag{33}$$

We now combine Eqs. (21), (22), (30), (31), having

$$\left(\sigma - \frac{\partial\psi_r(e)}{\partial e}[(1-B)m_0 + Bm_u]\right) : \delta e + (\psi_r(e)(m_0 - m_u) - E_B)\delta B = 0. \tag{34}$$

Let us follow a hypothetical experiment for lightly compacted materials. In this experiment we can guarantee that none of the particles will fracture, so that $\delta B = 0$, irrespective to the direction of the incremental strain path we are about to impose on the sample. Since the second term is now zero, the only way to satisfy Eq. (34) is by considering the following constitutive relation between the stress and the strain (which in this case is also the elastic strain):

$$\sigma = \frac{\partial\psi_r(e)}{\partial e}[(1-B)m_0 + Bm_u]. \tag{35}$$

We see that the elastic moduli are factorised, initially by m_0 when $B = 0$, but gradually by m_u. Since the first term in Eq. (34) is zero, and since $E_B\delta B$ (the dissipation) may be greater

than zero, the only way to satisfy Eq. (34), is to further consider the following relation:

$$\begin{aligned} E_B &= \psi_r(e)(m_0 - m_u) \\ &= \int_{d_m}^{d_M} \hat{\psi}(\Delta, e_e)(p_0(\Delta) - p_u(\Delta))\,\mathrm{d}\Delta \\ &= \Psi_0 - \Psi_u. \end{aligned} \tag{36}$$

We may term E_B as the 'breakage energy', describing the total stored energy that can be released from the system during the fracturing from the beginning state of initial particle distribution to the final state of an ultimate grain size distribution. Since we expect the breakage to grow such that $\delta B \geqslant 0$, the breakage energy should be non-negative, i.e., $E_B \geqslant 0$, as follows from Eq. (31). For that to happen m_0 is expected to be larger than m_u:

$$m_0 \geqslant m_u. \tag{37}$$

4.3. The hypothesis of energy split and the energy split function

As mentioned earlier, m_0 and m_u are two physical model parameters. Their value depends on the energy split function $f_\psi(d)$, the initial grain size distribution $p_0(d)$ and ultimate grain size distribution $p_u(d)$, via Eqs. (27) and (28). The exact definition of $p_0(d)$ and $p_u(d)$ was dealt with before. Here, we explore the hypothesis of energy split between grain size fractions, and examine particular forms of the energy split function $f_\psi(d)$.

In a randomly compacted assembly of granular materials smaller particles have a smaller coordination number, i.e., on average they contact fewer adjacent particles. Individual small particles may even 'float' between the micro-voids and hence carry no forces, and not store energy. Individual larger particles, on the other hand, are almost guaranteed to contact their neighbours, owing to their higher surface area. We may safely conclude that probabilistically larger particles store more energy. A rather general expression could be given by adopting the following non-dimensional power function for any $n \geqslant 0$:

$$f_\psi(d) = \left(\frac{d}{d_r}\right)^n, \tag{38}$$

where d_r is necessary for dimensional consistency, and to satisfy the requirement in Eq. (25). Based on Eqs. (20), (27) and (28):

$$m_0 = \langle d^n \rangle_0 d_r^{-n} = J_{n0} d_r^{-n}, \tag{39}$$

$$m_u = \langle d^n \rangle_u d_r^{-n} = J_{nu} d_r^{-n}, \tag{40}$$

where J_{n0} and J_{nu} are the nth order moment of the initial and ultimate grain size distributions. It is convenient to set the reference grain size according to

$$d_r = \sqrt[n]{J_{n0}} = \sqrt[n]{\langle d^n \rangle_0}. \tag{41}$$

Then

$$m_0 = 1 \tag{42}$$

which reduces Eq. (29) by

$$\psi_r(e) = \Psi(B \equiv 0). \tag{43}$$

Therefore, the physically measured strain energy along the boundaries of a representative volume element (i.e., the gross average stored energy) equates to the stored energy within the reference grain size of Eq. (41).

In these conditions

$$m_u = J_{nu}/J_{n0}. \tag{44}$$

Since bigger particles store more energy (as Eq. (38) suggests), the principle of energy minimisation requires that particles will get smaller. Since the ultimate grain size distribution is more dispersed towards smaller particles than the initial distribution, the ratio of their nth order moments is less than unity, i.e., satisfying our anticipation from Eq. (37).

The question is which value to assign for n? A simple model would be to postulate that the stored energy is proportional to the average projected area of particles, or otherwise the surface area of the particles. This is because larger particles attract more contact points linearly with increase in surface area, and more contact points means that more force chains will cross the particle. For an average force vector, this suggests a linear dependency between the stored energy and the surface area of the particle.

According to this model, for *a system of disks* the energy split function will simply be

$$f_\psi(d) = d/\overline{d}_0 \tag{45}$$

giving $m_0 = 1$ and $m_u = \overline{d}_u/\overline{d}_0$. For an ultimate fractal distribution, the average particle size $\overline{d}_u$ may be calculated using Eq. (A.10). Since particles get smaller, $m_0 \geqslant m_u$, guaranteeing a non-negative E_B and the continuous growth of breakage towards unity.

In *a system of spheres*, a linear dependency of stored energy on surface area gives

$$f_\psi(d) = d^2/J_{20} \tag{46}$$

and $m_0 = 1$ and $m_u = J_{2u}/J_{20}$. For an ultimate fractal distribution, the second order moment J_{2u} may be calculated using Eq. (A.11), while J_{20} would depend on the initial distribution.

We would now like to examine the hypothesis of energy split, as represented in Eq. (24). For that purpose, a simple series of numerical tests is presented using the DEM software by Itasca Consulting Group (2005) named pfc^{2d}. In this experiment, we would like to test the assumption in two extreme scenarios of poorly graded and well-dispersed samples. The first extreme situation is configured by setting different types of bimodal grain size distributions where the diameter ratios between the big and small particles and their numbers alters in three different tests. The second more disperse case, examines a uniform grain size distribution by number (rather than by mass), where the ratio between the biggest and smallest particles in the sample is 10, and this test is repeated three times. In all experiments, the particles are thrown into a box, then let to redistribute the unbalanced internal contact forces, till they settle quietly at a target overall porosity that ensures they will be tightly compacted. Fig. 6 presents the force chain distribution between the particles in two typical cases. The left figure describes the result for a bi-modal distribution with $d_{\text{large}} = 10d_{\text{small}}$. Clearly, the bigger particles attract many more force chains than the smaller particles due to the larger surface area and more contacting points. This strengthens the energy split hypothesis, and relates to the 'cushioning effect' described in Fig. 2.

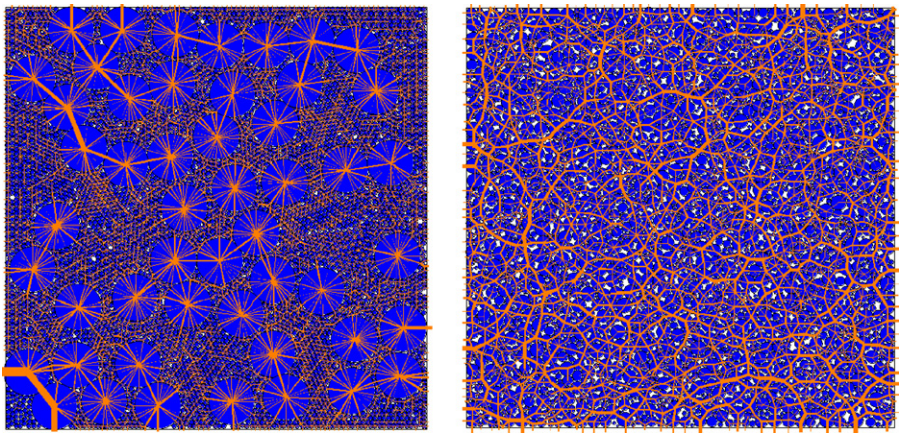


Fig. 6. Force chains in compacted granular assemblies using DEM: (left) bi-modal distribution ($d_{large} = 10d_{small}$) and (right) uniform distribution (by size).

The right picture in Fig. 6 presents a standard force chain distribution in well-graded materials, making the examination of the energy split hypothesis harder. For that purpose it is possible to calculate the stored energy associated with each of the particles in the test. Let us denote all of the particles with a size d in the fraction $d - \delta d/2 < d < d + \delta d/2$ in the representative random sample uniformly by a. Since we have employed a linear contact-stiffness model, the stored energy within all of the particles a is calculated via summation:

$$\psi_a(d) = \frac{1}{2}\sum_{i=1}^{N_c}(|Q_i^N|^2/k^N + |Q_i^S|^2/k^S), \tag{47}$$

where N_c is the number of contact points; $|Q_i^N|$ and $|Q_i^S|$ are the magnitudes of the normal and shear components of the contact force; and k^N and k^S are the normal and shear-contact stiffnesses. In the bi-modal problem, there are only two values calculated based on Eq. (47). In the uniform distribution case, there are essentially infinite fractions. Therefore, since our sample is limited by the amount of grains, we take $\delta d = (d_{\text{large}} - d_{\text{small}})/20$. The average stored energy $\hat{\psi}(d, e)$ for particles with the size d may then be estimated by the summation over the representative volume:

$$\hat{\psi}(d, e) = \sum_a \psi_a(d)/N_a, \tag{48}$$

where N_a is the total number of particles in the fraction. The energy split function can therefore be estimated in each of the experiments by combining the last expression with Eq. (24):

$$f_\psi(d) = \frac{\sum_a \psi_a(d)/N_a}{\psi_r(e)} = \frac{N_r}{N_a}\frac{\sum_a \psi_a(d)}{\sum_r \psi_r(d_r)}, \tag{49}$$

where r denote particles with a reference grain size d_r.

Fig. 7 plots the numerical values for the various fractions in the corresponding tests. In this figure, the simple theoretical expression for the system of disks is plotted via Eq. (45). The agreement between this expression and the numerical results is remarkable,

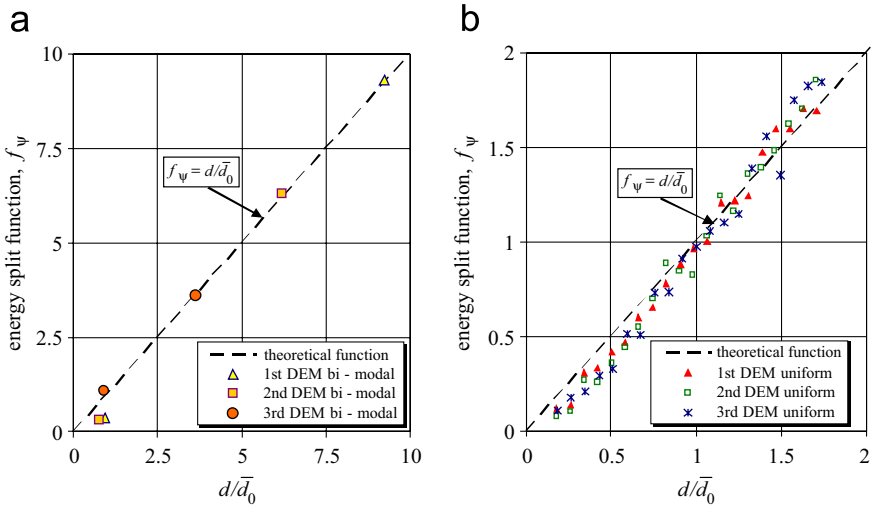


Fig. 7. Comparison between numerical values and the simple theoretical expression of the energy split function for disk assemblies, based on (a) bi-modal and (b) uniform distributions.

considering the simplicity of the model. It is clear that larger particles store more energy than smaller particles, and that this relation is nearly linear, if not slightly exponential with growing diameters.

4.4. The residual breakage energy

Based on Eq. (10) it is possible to rephrase Eq. (36):

$$E_B = \frac{1}{B}(\Psi_0 - \Psi) \tag{50}$$

which also suggests an alternative definition of breakage, as being the relative stored energy that has been released from the system

$$B = \frac{\Psi_0 - \Psi}{\Psi_0 - \Psi_u}. \tag{51}$$

Initially, the energy is completely stored in the unbroken particles, such that $\Psi = \Psi_0$ and $B = 0$. However, as the particles are crushed, the stored energy tends to $\Psi = \Psi_u$, and $B = 1$.

It is also useful to write the following expression:

$$1 - B = \frac{\Psi - \Psi_u}{\Psi_0 - \Psi_u} \tag{52}$$

representing the residual breakage, or the relative stored energy that remains in the system. Finally, we define the 'residual breakage energy' that is available in the system for crushing grains:

$$E_B^* = E_B(1 - B) = \Psi - \Psi_u. \tag{53}$$

The incremental change of the residual breakage energy may be expressed by

$$\delta E_B^* = \delta E_B(1-B) - E_B\delta B. \tag{54}$$

5. Yielding and fracture in CBM models

5.1. Yield condition

Based on Eq. (31) we define the breakage yield function $y_B = y_B(E_B)$:

$$\lambda_B y_B(B, E_B) \equiv E_B\delta B - \tilde{\Phi}_B(B, \delta B) = 0. \tag{55}$$

This equation is a degenerate special case of Legendre transformation for first order homogeneous functions, whereby $\lambda_B \geqslant 0$ denotes the non-negative breakage multiplier. We see that in order to satisfy the equality, the yield function must be conditioned by $y_B(E_B) \leqslant 0$. Differentiating both sides by E_B gives the evolution law for the breakage, as an associated flow rule to $y_B(E_B) = 0$:

$$\delta B = \lambda_B \frac{\partial y_B(B, E_B)}{\partial E_B}. \tag{56}$$

5.2. Postulate of breakage growth criterion

Let us postulate the following yield criterion[1]:

$$y_B(B, E_B) = E_B^*(1-B) - G_B = E_B(1-B)^2 - G_B \leqslant 0, \tag{57}$$

where G_B is a strain energy constant of material. The breakage dissipation may be written explicitly by using the equality in (57) and substituting in (31) as

$$\tilde{\Phi}_B = E_B\delta B = G_B(1-B)^{-2}\delta B. \tag{58}$$

Once yielding occurs then the consistency condition ($\delta y_B = 0$) requires that $\delta E_B(1-B)^2 - 2E_B(1-B)\delta B = 0$, i.e., using Eq. (54):

$$\tilde{\Phi}_B = \delta E_B^* \tag{59a}$$

which can be interpreted as follows:

energy dissipation from breakage = loss in residual breakage energy. (59b)

As Eq. (53) suggests, the residual breakage energy E_B^* is linked to the area between the current and ultimate grain size cumulative curves (see Fig. 8). The change in the residual breakage energy δE_B^* for a given incremental increase of breakage δB, is in fact linked to the area captured between the current curve and the 'new' current curve attained immediately after imposing the increment of breakage. By adopting the consistency condition of the postulated breakage yield criterion, we are in fact assuming that the energy dissipation from breakage $\tilde{\Phi}_B = E_B\delta B$ is simply given by the change in the residual breakage energy δE_B^*. In other words, the breakage dissipation is linked to the area change of the grain size distribution from moving the current distribution function to a new

[1]We emphasise that 'yield criterion' is only a subset of possible 'yield conditions' in Eq. (55), so much as von Mises or Tresca criteria are only two possibilities of the entire family of elastoplastic yield conditions.

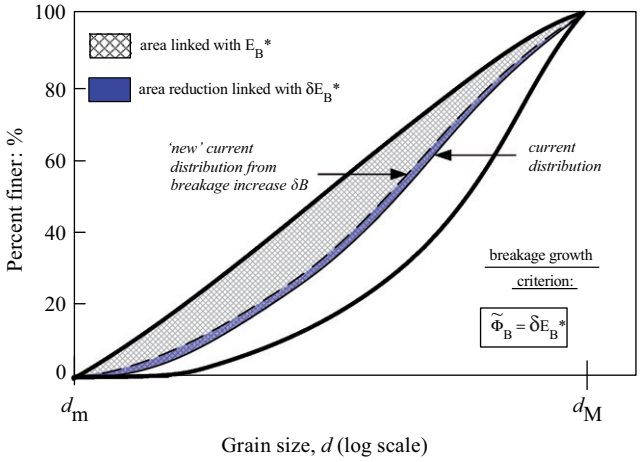


Fig. 8. Schematic representation of the physical meaning of the postulated breakage growth criterion.

position. This proposition is very attractive since this area change relates only to those particles that undergo crushing and dissipated energy during this time step.

Combining Eqs. (57) and (58) gives the incremental growth of breakage in this situation:

$$\delta B = \lambda_B (1 - B)^2 \tag{60}$$

which gives

$$\lambda_B = \frac{\tilde{\Phi}_B}{G_B}, \tag{61}$$

$$\delta B = (1 - B)^2 \frac{\tilde{\Phi}_B}{G_B} = \frac{\tilde{\Phi}_B}{E_B}. \tag{62}$$

The non-negative breakage multiplier can therefore be interpreted as a normalised rate of change of breakage dissipation. The incremental change of breakage is the ratio between the breakage dissipation and the breakage energy. In terms of the field variables, the increment of breakage may be derived directly by manipulating the consistency condition:

$$\delta B = \frac{(1 - B)}{2E_B} \delta E_B. \tag{63}$$

The incremental change of the breakage energy may be expressed by manipulating Eqs. (22), (35) and (36):

$$\delta E_B = \frac{\vartheta}{1 - \vartheta B} \tilde{W}. \tag{64}$$

Thus, the amount of work which is being dissipated from the system may be expressed by

$$\tilde{\Phi}_B = \frac{\vartheta}{2} \frac{1 - B}{1 - \vartheta B} \tilde{W} \tag{65}$$

whereby we designate

$$\vartheta = 1 - m_u = 1 - J_{nu}/J_{n0} \tag{66}$$

noting Eqs. (42) and (44). In elastic-breakage models, the total work ($\tilde{W} = \sigma : \delta e$) is also the 'elastic work' ($\sigma : \delta e_e$). Initially a factor of $\vartheta/2$ of the work ($\tilde{W} = \sigma : \delta e$) is being dissipated, but as breakage proceeds, the dissipation from breakage is gradually stopping and all of the work is being stored elastically in the system.

The breakage energy is initially given by Eq. (36), but once the equality in the yielding criterion (57) is met, the breakage may be expressed as a function of the breakage energy:

$$B = 1 - \left(\frac{G_B}{E_B}\right)^{1/2}. \tag{67}$$

During initial yielding, the breakage energy will be $E_B = G_B$ which corresponds to $B = 0$. As the breakage proceeds, the breakage energy increases and $B \to 1$, and therefore the postulated yield criterion would always limit breakage as required by Eq. (9). As breakage grows the breakage energy which is needed for yielding increases. Since the breakage energy is related to the amount of stored energy in the system, this aspect explains the emergence of what is normally termed isotropic hardening in critical state soil mechanics. This aspect is explored in the second part of the paper (Einav, 2006), where various models are developed based on the current uncoupled elastic-breakage theory and further advancements that set a richer framework that is coupled with plasticity.

6. Conclusions

A new continuum mechanics theory for the constitutive modelling of brittle granular matter has been proposed. We start by advancing the concept of breakage based on the existence of an ultimate grain size distribution. Breakage essentially measures the relative proximity of the current grain size distribution to the initial and ultimate distributions. Therefore, the breakage is confined to increase from zero to one with the increase in surface area, exactly as damage is when applied in continuum damage mechanics (CDM) models. We term the new theory continuum breakage mechanics (CBM), reflecting the analogy with CDM. However, while damage effectively represents the opening of micro-cavities and cracks in solid-like materials, breakage represents comminution of particles in discrete granular media, for which damage is simply ambiguous. The two measures of breakage and damage may therefore be seen as complementary, but dissimilar. Whereas damage is generally insensitive to compression and increases in extension, the breakage is active in compression but insensitive to extension.

The incorporation of breakage in continuum models is made possible by adopting a statistical homogenisation procedure in which the grain size distribution is used as a weight function of the stored energy in the different grain fractions. By following a thermomechanical analysis, we identify the thermodynamic couple of breakage in the form of the 'breakage energy'. We show that this property is always positive by hypothesising that larger particles store more energy than smaller particles. This hypothesis, which is tested numerically in the paper, may explain the fact that particles become smaller with time in accordance with the principle of energy minimisation. The rate at which the energy is dissipated is postulated to be linked with the rate of change of the 'residual breakage energy' that represents the amount of stored energy that is available for release from the system. The incremental change in residual breakage energy is related with the area confined between the current and 'new' current particle size distributions,

associated only with those particles that undergo crushing. It is shown that this postulate provides an explanation of the so-called isotropic hardening in granular materials, in a direct physical way, without the imposition of phenomenological curve-fitting of compression curves as typically used in critical state soil mechanics. To derive predictive models, it is important to account for additional dissipative mechanisms that arise from frictional plasticity. This is discussed in part II of this paper (Einav, 2006), in which the current elastic-breakage theory is enhanced by coupling it with plasticity.

The CBM Theory shows that in brittle granular materials isotropic hardening is related to grain crushing and this is similar to the conclusion obtained based on the clastic hardening approach of McDowell et al. (1996). The arguments in the clastic hardening approach are rooted in the statistics of a single grain crushing and to the consequential increase in surface area. While single grain crushing should have influence on the assembly behaviour, it is the constraining comminution effect—the cushioning effect—which governs the constitutive behaviour of the granular material to the first order. The consequences of this are that larger grains store more elastic energy than smaller particles, a fact which is fundamental to the current analysis. This bases the CBM Theory on the overall assembly crushing statistics while using the grain size distribution and how it evolves with breakage growth. While the clastic hardening approach is limited to one-dimensional compression loading conditions, the CBM Theory allows for any loading direction to be accounted for by adopting modern thermodynamics procedures.

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Appendix A. Deriving fractal distribution by mass

In a fractal granular medium the number of the collection of particles that have diameters Δ above the size d is expressed by (e.g., Turcotte, 1986)

$$N(\Delta > d) = Cd^{-\alpha}, \tag{A.1}$$

where C is a constant of proportionality and α is the fractal dimension. However, in a conventional sieve analysis, the mass of all of the particles that are finer than the sieve mesh size d is normally measured, i.e.,

$$M_d(\Delta < d) = \int_{d_m}^{d} s\rho\Delta^3 \, \mathrm{d}N(\Delta), \tag{A.2}$$

where s denotes the shape factor, which for spheres becomes $s = \pi/6$, and ρ is the specific mass, both assumed constant for the different fractions. The number of particles in a fraction Δ, $\mathrm{d}N(\Delta)$ may be given by differentiating Eq. (A.1) (e.g., McDowell et al., 1996):

$$\mathrm{d}N(\Delta) = C\alpha\Delta^{-\alpha-1} \, \mathrm{d}\Delta \tag{A.3}$$

so that

$$M_d(\Delta<d)=\frac{C\alpha s\rho}{3-\alpha}(d^{3-\alpha}-d_m^{3-\alpha}). \tag{A.4}$$

The total mass of the sample is

$$M_T=M_d(\Delta<d_M)=\frac{C\alpha s\rho}{3-\alpha}(d_M^{3-\alpha}-d_m^{3-\alpha}) \tag{A.5}$$

such that based on the fractal model, the grain size cumulative distribution *by mass* is

$$F(d)=\frac{M_d(\Delta\leqslant d)}{M_T}=\frac{d^{3-\alpha}-d_m^{3-\alpha}}{d_M^{3-\alpha}-d_m^{3-\alpha}}. \tag{A.6}$$

It is interesting to notice that this distribution has a limit when $\alpha\to 3$:

$$F(d)=\frac{\log(d/d_m)}{\log(d_M/d_m)}. \tag{A.7}$$

If d_m was set to zero, this limit would not exist. However, for most practical purposes, α is around 2.6 as discussed in the text, and d_m could indeed be assumed zero. In this case, the result in Eq. (A.6) is identical to the one obtained by Turcotte (1986):

$$F(d)=\left(\frac{d}{d_M}\right)^{3-\alpha}. \tag{A.8}$$

The grain size distribution is given by differentiating $F(d)$:

$$p(d)=d_M^{-1}(3-\alpha)\left(\frac{d}{d_M}\right)^{2-\alpha}. \tag{A.9}$$

The first order moment of this distribution, or the averaged particle size, is given by using Eq. (18):

$$\overline{d}=\frac{3-\alpha}{4-\alpha}d_M. \tag{A.10}$$

The second order moment of the grain size distribution $p(d)$ is given by applying Eq. (19):

$$J_2=\frac{3-\alpha}{5-\alpha}d_M^2. \tag{A.11}$$

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